

Practical 10-

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

CODE-

```
GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int visited[8] = {0};
    int head = 53;
    int total = 0;

    for(int i = 0; i < 8; i++) {
        int min = 9999, index = -1;

        for(int j = 0; j < 8; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);
                if(dist < min) {
                    min = dist;
                    index = j;
                }
            }
        }

        total += min;
        head = req[index];
        visited[index] = 1;

        printf("Move to %d (Distance = %d)\n", head, min);
    }

    printf("Total Head Movement = %d\n", total);
    return 0;
}
```

OUTPUT-

```
HP@DESKTOP-B4BD73N MINGW64 ~  
$ nano sstf.c  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ gcc sstf.c -o sstf  
./sstf  
Move to 41 (Distance = 12)  
Move to 65 (Distance = 24)  
Move to 67 (Distance = 2)  
Move to 98 (Distance = 31)  
Move to 122 (Distance = 24)  
Move to 124 (Distance = 2)  
Move to 183 (Distance = 59)  
Move to 14 (Distance = 169)  
Total Head Movement = 323
```